

Analysis group meeting

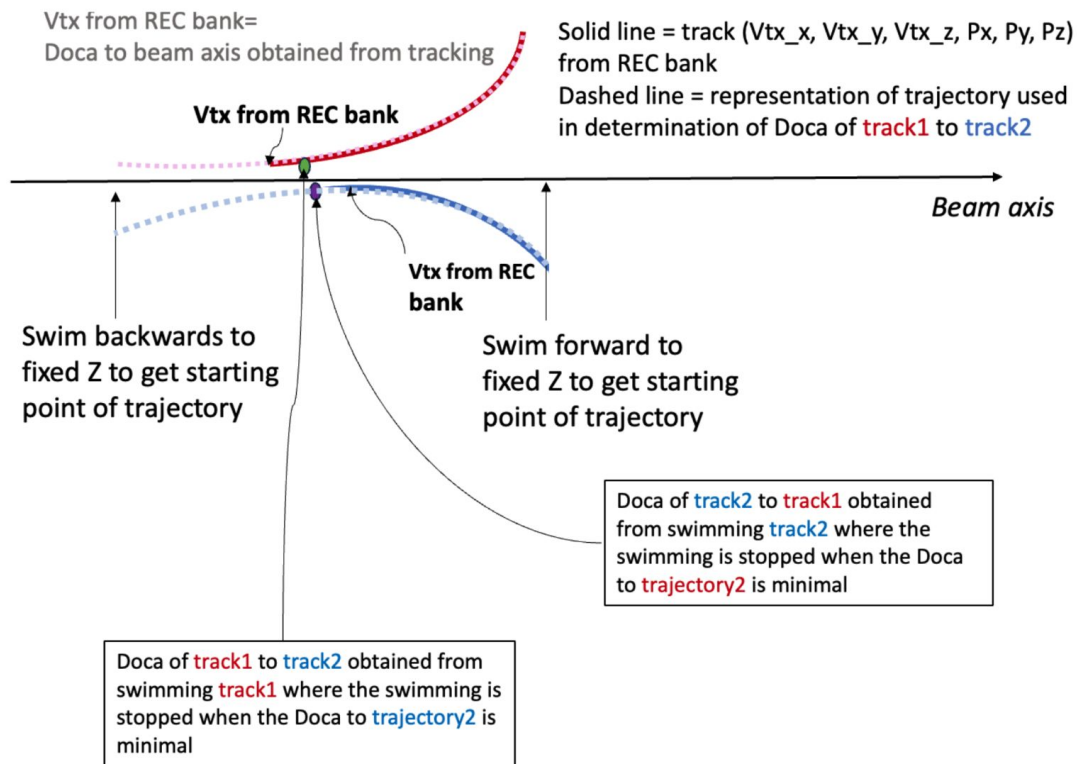
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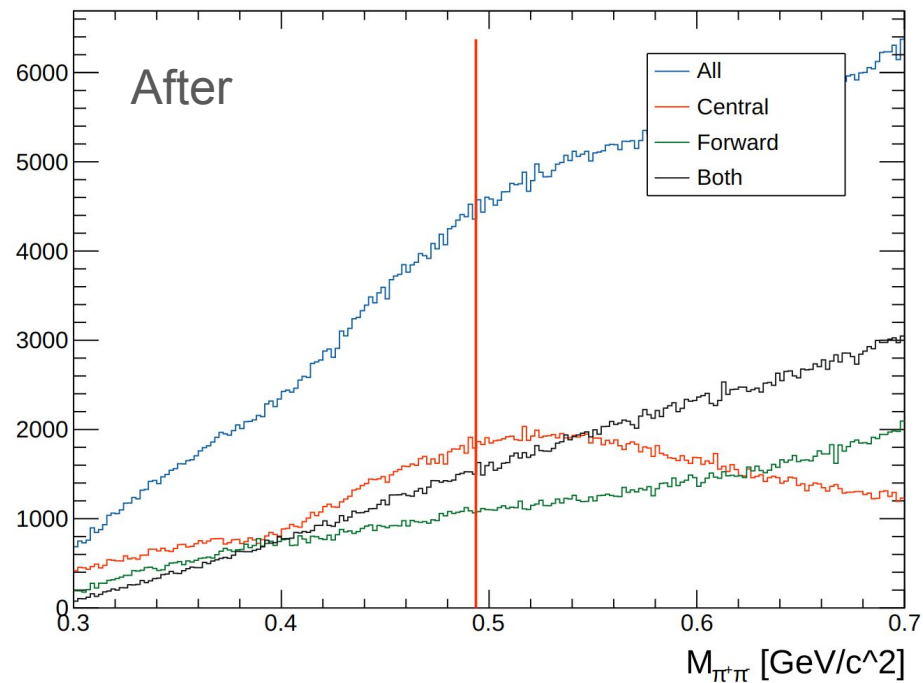
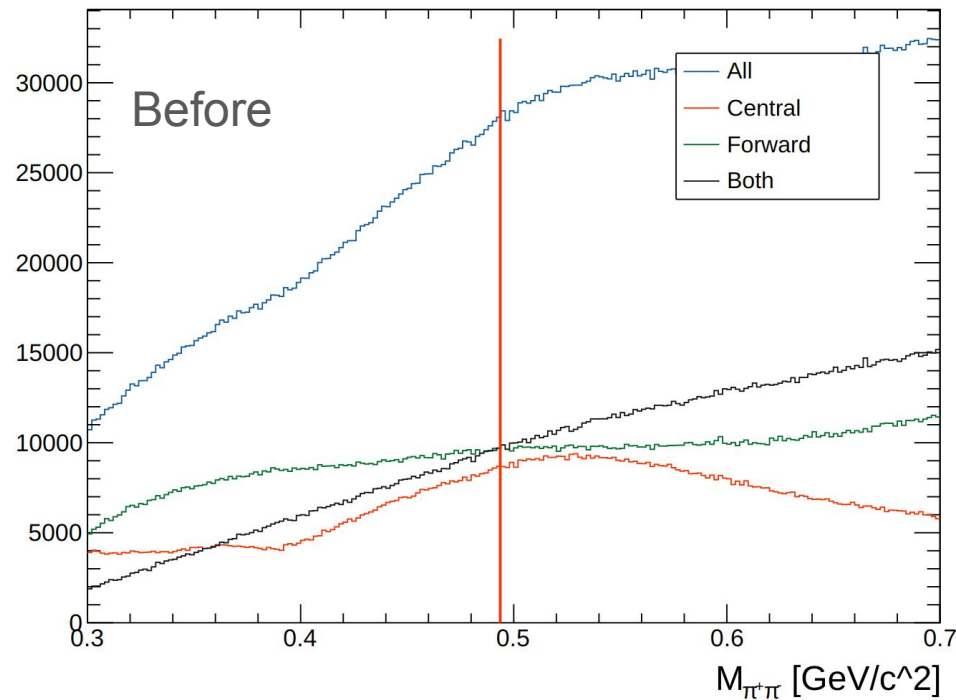
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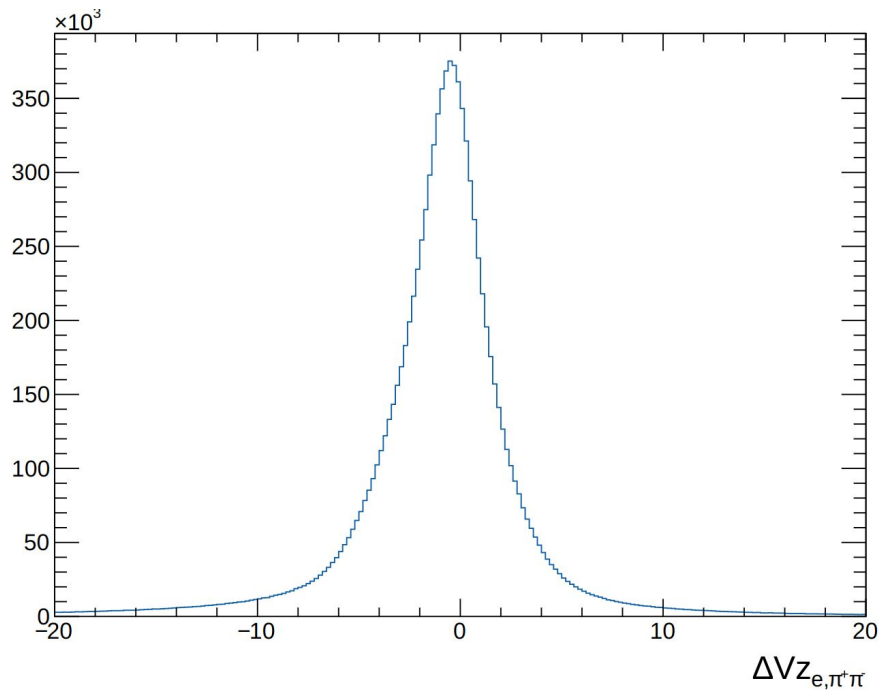
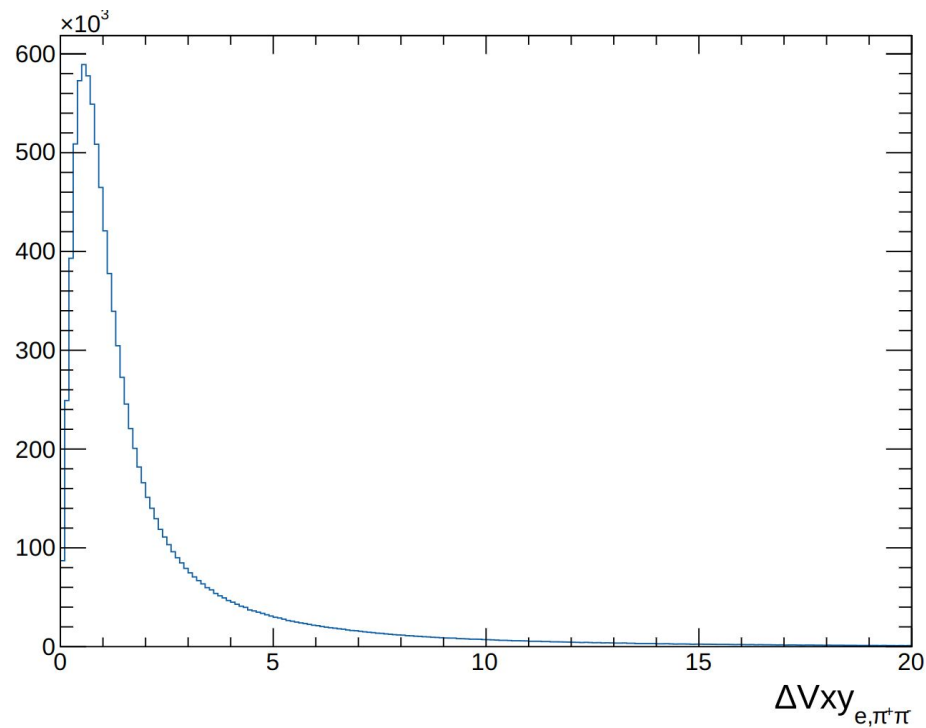
Detached vertex study



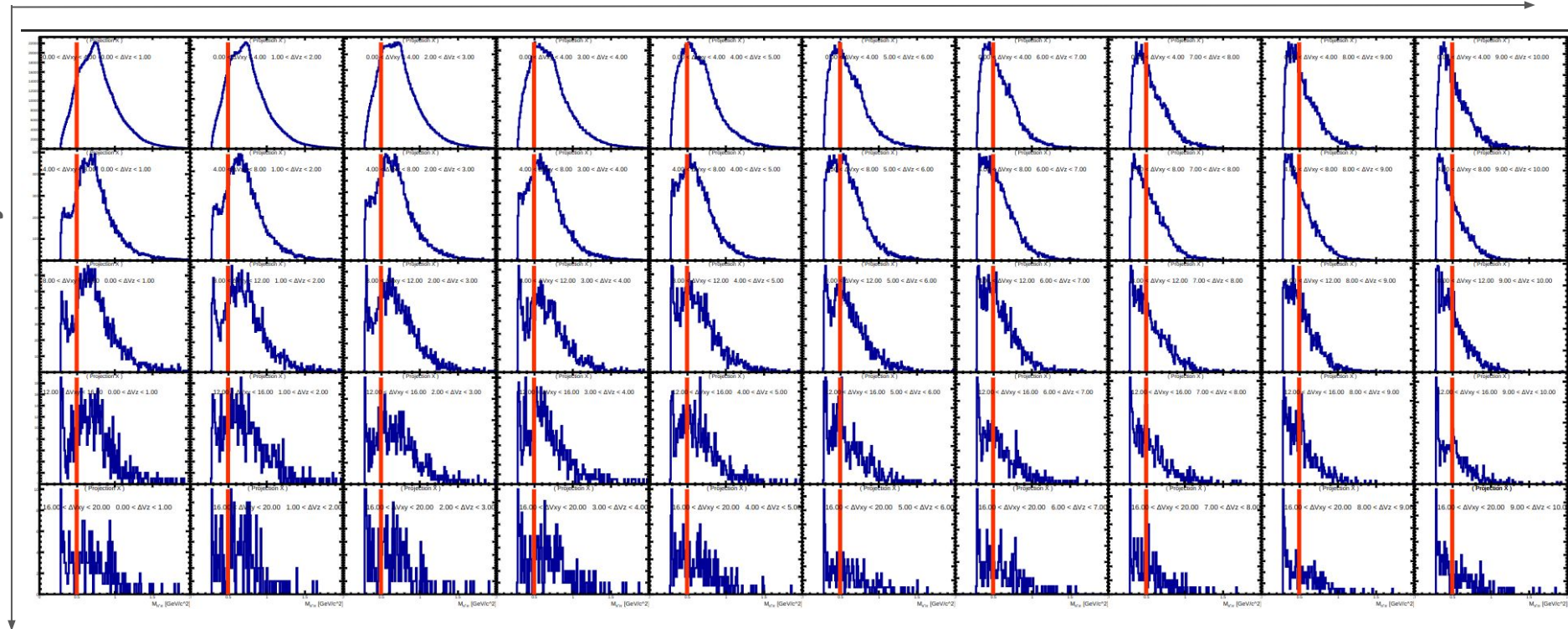
- Cuts electron:
 - $-10 < V_z < 0$ cm
 - $-3 < \chi^2_{pid} < 3$
- Pions:
 - $-10 < V_z < 0$ cm
 - $-10 < \chi^2_{pid} < 10$
- Kinematical cuts:
 - $Q^2 > 1 \text{ GeV}^2$
 - $W > 2 \text{ GeV}$
 - $0 < z_h < 1$
- Detached vertex cuts:
 - $r < 5 \text{ mm}$
 - pions vertex to be downstream of the e- vertex

Use LD2 data from Pass0v7





Detached vertex study

 ΔV_z
 ΔV_{xy}


Use Pass0v7 data, select event with a trigger electron with $\text{pid}=11$ and $\text{status}<0$

Charged pions identified with $\text{pid}=211$ and $\text{pid}=-211$.

Strategy:

Plot V_x and V_y distribution for the electron, fit to get mean and sigma, cut at 11 sigma

Plot V_z distribution for the electron, fit to get mean and sigma for the entrance/exit of the LD2 target and the mean and sigma for the CxC or CuSn targets, cut a 3 sigma to remove contamination for the LD2 target and separate Cu and Sn target.

Plot V_x and V_y distribution for the pion, fit to get mean and sigma, cut at 11 sigma

Plot V_z distribution for the pion, fit to get 2 mean and sigma for the CxC target or 1 mean and sigma for the Cu and Sn target

RG-D: Vertex study: Carbon

Electron	V _x		V _y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.003076	0.215686	-0.164755	0.229661
Sector 1	0.003140	0.060235	-0.329608	0.277007
Sector 2	0.287538	0.311189	-0.279691	0.143903
Sector 3	0.053724	0.235268	-0.152139	0.185990
Sector 4	-0.031792	0.104016	0.233189	0.319917
Sector 5	-0.255993	0.286280	-0.069730	0.148932
Sector 6	-0.026808	0.230069	-0.182212	0.189967

Electron	V _z	
	μ [cm]	σ [cm]
Entrance of the empty LD2 cell	-18.405000	1.190936
Exit of the empty LD2 cell	-13.831574	1.045480
first foil	-8.485692	0.711907
second foil	-3.589993	0.817241

RG-D: Vertex study: Carbon

π^+	V_x		V_y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.012299	0.180807	-0.178737	0.144222
Sector 1	-0.078470	0.131502	-0.437228	0.378782
Sector 2	0.100359	0.239559	-0.382100	0.281367
Sector 3	-0.182316	0.392859	-0.220538	0.129758
Sector 4	0.051276	0.105245	0.122445	0.450105
Sector 5	-0.107038	0.233748	0.027162	0.288730
Sector 6	0.105968	0.357065	-0.133877	0.117619

π^-	V_x		V_y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.005448	0.254900	-0.169730	0.244652
Sector 1	0.083143	0.175538	-0.467380	0.402852
Sector 2	0.461195	0.443460	-0.222087	0.172187
Sector 3	0.047715	0.236601	-0.138140	0.314320
Sector 4	-0.197082	0.240548	0.422439	0.461624
Sector 5	-0.349982	0.410601	-0.122299	0.156765
Sector 6	-0.004608	0.221541	-0.148201	0.312715

π^+	V_z	
	μ [cm]	σ [cm]
first foil	-7.575302	0.734257
second foil	-2.636661	0.655775

π^-	V_z	
	μ [cm]	σ [cm]
first foil	-8.337641	0.731892
second foil	-3.495834	0.849354

Electron	V_x		V_y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.002885	0.217284	-0.165301	0.232290
Sector 1	0.001914	0.059183	-0.324606	0.280696
Sector 2	0.283940	0.313940	-0.283482	0.144185
Sector 3	0.052292	0.239170	-0.155345	0.187858
Sector 4	-0.030355	0.104842	0.239413	0.324523
Sector 5	-0.256868	0.286190	-0.068250	0.150656
Sector 6	-0.021374	0.232851	-0.179737	0.188914

Electron	V_z	
	μ [cm]	σ [cm]
Entrance of the empty LD2 cell	0.000000	0.000000
Exit of the empty LD2 cell	0.000000	0.000000
first foil	-8.537621	1.464062
second foil	-2.615931	3.033578

π^+	V_x		V_y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.011423	0.176051	-0.184492	0.131330
Sector 1	-0.074644	0.130805	-0.443186	0.388910
Sector 2	0.090227	0.235431	-0.381319	0.284524
Sector 3	-0.194080	0.403671	-0.224642	0.116495
Sector 4	0.051959	0.107386	0.116433	0.450430
Sector 5	-0.098382	0.231788	0.013685	0.280017
Sector 6	0.117976	0.362657	-0.143893	0.109511

π^+	V_z	
	μ [cm]	σ [cm]
first foil	-7.636740	0.725776
second foil	0.104842	0.000000

π^-	V_x		V_y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.006150	0.255690	-0.170247	0.246203
Sector 1	0.087344	0.183259	-0.465011	0.405056
Sector 2	0.466924	0.454457	-0.218707	0.172498
Sector 3	0.047479	0.237066	-0.137766	0.323342
Sector 4	-0.207361	0.247520	0.418120	0.466678
Sector 5	-0.354526	0.421127	-0.128478	0.156677
Sector 6	-0.003711	0.220499	-0.145897	0.322454

π^-	V_z	
	μ [cm]	σ [cm]
first foil	-8.438102	0.787356
second foil	0.000000	0.056862

Electron	V _x		V _y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.002885	0.217284	-0.165301	0.232290
Sector 1	0.001914	0.059183	-0.324606	0.280696
Sector 2	0.283940	0.313940	-0.283482	0.144185
Sector 3	0.052292	0.239170	-0.155345	0.187858
Sector 4	-0.030355	0.104842	0.239413	0.324523
Sector 5	-0.256868	0.286190	-0.068250	0.150656
Sector 6	-0.021374	0.232851	-0.179737	0.188914

Electron	V _z	
	μ [cm]	σ [cm]
Entrance of the empty LD2 cell	0.000000	0.000000
Exit of the empty LD2 cell	0.000000	0.000000
first foil	-8.537621	1.464062
second foil	-2.615931	3.033578

RG-D: Vertex study: Sn

π^+	V _x		V _y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.011742	0.178725	-0.184621	0.129364
Sector 1	-0.082659	0.137204	-0.455065	0.387410
Sector 2	0.095771	0.236690	-0.389291	0.285450
Sector 3	-0.180844	0.400945	-0.224270	0.115646
Sector 4	0.056528	0.111261	0.128826	0.451400
Sector 5	-0.102635	0.232835	0.021405	0.284308
Sector 6	0.115379	0.357011	-0.143841	0.107755

π^+	V _z	
	μ [cm]	σ [cm]
first foil	-2.689639	0.741223
second foil	0.104842	0.000000

π^-	V _x		V _y	
	μ [cm]	σ [cm]	μ [cm]	σ [cm]
Sector independant	-0.003436	0.257639	-0.169969	0.245218
Sector 1	0.085921	0.182208	-0.474633	0.414718
Sector 2	0.466683	0.453141	-0.221688	0.172840
Sector 3	0.048474	0.238822	-0.136937	0.323201
Sector 4	-0.205547	0.248984	0.434893	0.473129
Sector 5	-0.355318	0.415367	-0.124467	0.156805
Sector 6	0.000286	0.223391	-0.138792	0.318648

π^-	V _z	
	μ [cm]	σ [cm]
first foil	-3.501254	0.776460
second foil	0.000000	0.018749